Review

Linear combinations

Suppose we are given vectors $\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_p$ in $\mathbb{R}^n$, and scalars $c_1, ..., c_p$. The vector

$$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p$$

is called a linear combination of $\mathbf{v}_1, ..., \mathbf{v}_p$ with weights $c_1, ..., c_p$.

Span$\{\mathbf{v}_1, ..., \mathbf{v}_p\}$

- The set of all linear combinations of $\mathbf{v}_1, ..., \mathbf{v}_p$ is called the subset of $\mathbb{R}^n$ spanned (or generated) by $\mathbf{v}_1, ..., \mathbf{v}_p$.

This subset is represented by

$$\text{Span}\{\mathbf{v}_1, ..., \mathbf{v}_p\}$$

Review

Linear dependence/independence

Suppose $\mathbf{v}_1, ..., \mathbf{v}_p$ are vectors in $\mathbb{R}^n$.

Consider the vector equation

$$x_1\mathbf{v}_1 + \cdots + x_p\mathbf{v}_p = \mathbf{0} \qquad (\star)$$

The equation $(\star)$ has a trivial solution $x_1 = \cdots = x_p = 0$.

- $\mathbf{v}_1, ..., \mathbf{v}_p$ are said to be linearly dependent if $(\star)$ has a nontrivial solution (i.e., a solution other than $x_1 = \cdots = x_p = 0$).

- $\mathbf{v}_1, ..., \mathbf{v}_p$ are said to be linearly independent if $(\star)$ does not have nontrivial solutions (i.e., its *unique* solution is $x_1 = \cdots = x_p = 0$).

Review

Example

The vectors $\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ -5 \\ 4 \end{bmatrix}$ are linearly dependent:

The equation

$$x_1 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ -5 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

has a nontrivial solution $x_1 = 1, x_2 = 2, x_3 = -1$.

$$\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ -5 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

weights 1,2,-1.

Review

Example

The vectors $\begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 2 \\ -4 \end{bmatrix}$ are linearly dependent:

The equation

$$x_1 \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

has a nontrivial solution $x_1 = 2, x_2 = 0, x_3 = 1$.

$$2 \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad \text{weights } 2, 0, 1$$

Review

Example

The vectors $\begin{bmatrix} 1 \\ -2 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ are linearly dependent:

The equation

$$x_1 \begin{bmatrix} 1 \\ -2 \\ -3 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

has a nontrivial solution $x_1 = 0, x_2 = 0, x_3 = -\frac{1}{5}$.

$$0 \begin{bmatrix} 1 \\ -2 \\ -3 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{5} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

weights $0, 0, -\frac{1}{5}$

Suppose $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are vectors in $\mathbb{R}^n$. If one of the $\mathbf{v}_i$'s is zero, then $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are linearly dependent.

Review

Problem. Given a list of vectors $\mathbf{a}_1, ..., \mathbf{a}_p$ in $\mathbb{R}^n$, determine if they are linearly dependent or linearly independent.

Solution. You should solve the vector equation

$$x_1\mathbf{a}_1 + \cdots + x_p\mathbf{a}_p = \mathbf{0}$$

or equivalently the linear system with augmented matrix

$$A = [\ \mathbf{a}_1\ \cdots\ \mathbf{a}_p \mid \mathbf{0}\ ]$$

- The system has a nontrivial solution $\Rightarrow$ The vectors are linearly dependent.
- The system does not have any nontrivial solutions $\Rightarrow$ the vectors are linearly independent.

• A homogeneous linear system has nontrivial solutions if and only if it has free variables.

Example

Are the columns of the following matrix linearly dependent or independent?

$$A = \begin{bmatrix} 3 & -1 & 1 \\ 3 & 1 & 0 \\ 6 & -2 & -5 \end{bmatrix}$$

Solution.

- We should determine if the linear system with the following augmented matrix has a nontrivial solution :

$$[\, A \,|\, \mathbf{0} \,] = \left[\begin{array}{ccc|c} 3 & -1 & 1 & 0 \\ 3 & 1 & 0 & 0 \\ 6 & -2 & -5 & 0 \end{array}\right]$$

- We row reduce this matrix to obtain an E.F.

$$\left[\begin{array}{ccc|c} 3 & -1 & 1 & 0 \\ 3 & 1 & 0 & 0 \\ 6 & -2 & -5 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 3 & -1 & 1 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & -7 & 0 \end{array}\right] \quad \text{E.F.}$$

- There are no free variables, i.e., the system has no nontrivial solutions.
- It follows that the columns of A are **linearly independent**.

Theorem

Suppose $\mathbf{v}_1, ..., \mathbf{v}_p$ are vectors in $\mathbb{R}^n$. If $p > n$, then $\mathbf{v}_1, ..., \mathbf{v}_p$ are linearly dependent.

Example. Are the vectors

$$\begin{bmatrix} -1 \\ 2 \\ 1 \\ -5 \end{bmatrix}, \begin{bmatrix} -4 \\ 5 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 1 \\ 7 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \\ -6 \\ 4 \end{bmatrix} \text{ and } \begin{bmatrix} 1 \\ 6 \\ 2 \\ -5 \end{bmatrix}$$

linearly dependent or independent?

Answer. They are linearly dependent because we have 5 vectors in $\mathbb{R}^4$ and $5 > 4$!

The punchline...

If we choose more than n vectors in $\mathbb{R}^n$, then they will be linearly dependent.

Warning. This does not mean that if we choose n or fewer vectors in $\mathbb{R}^n$ then they will be linearly independent!

Example

Question. We are given 8 vectors in $\mathbb{R}^5$. Can they be linearly independent?

Answer. No, they can't, because $8 > 5$.

Question. If we are given 3 vectors in $\mathbb{R}^5$, are they always linearly independent?

Answer. No, for example, if we are given

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

then

$$1 \cdot \mathbf{v}_1 + 1 \cdot \mathbf{v}_2 + 1 \cdot \mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Therefore the above vectors are linearly dependent.

Entries of a matrix

- An $m \times n$ matrix A can be represented as

$$A = [\ \mathbf{a}_1\ \cdots\ \mathbf{a}_n\ ]$$

where $\mathbf{a}_1, ..., \mathbf{a}_n$ are vectors in $\mathbb{R}^m$.

- Sometimes we represent A as

$$A = [a_{ij}]$$

where a_{ij} is called the (i,j)-entry and is the entry in the i-th row and j-th column.

Exmaple.
Suppose $A = \begin{bmatrix} 1 & -1 & 4 & 0 & 5 \\ 2 & -\frac{1}{4} & 3 & -3 & 1 \\ -3 & \frac{3}{2} & -\sqrt{3} & 0 & 0 \end{bmatrix}$.

Then :

- The (1,3)-entry of A is 4. We can also say $a_{1,3} = 4$.
- The (3,2)-entry of A is $\frac{3}{2}$. We can also say $a_{3,2} = \frac{3}{2}$.
- The (2,5)-enty of A is 1. We can also say $a_{2,5} = 1$.

Terminology

- diagonal entries of a matrix are the (i,i)-entries for various i:

  The diagonal entries of $\begin{bmatrix} 1 & -1 & 4 & 0 & 5 \\ 2 & -\frac{1}{4} & 3 & -3 & 1 \\ -3 & \frac{3}{2} & -\sqrt{3} & 0 & 0 \end{bmatrix}$ are $1, -\frac{1}{4}, -\sqrt{3}$.

- A square matrix is a matrix of size $m \times m$ (i.e. the same number of rows as columns).

- A diagonal matrix is a square matrix whose off-diagonal entries are zero : $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{3} \end{bmatrix}$ is a diagonal matrix.

- An $m \times n$ matrix in which all entries are equal to zero is called a zero matrix. Usually we write a zero matrix as $0_{m\times n}$ or just 0.

Terminology (cont.)

- The $n \times n$ identity matrix, which is usually denoted by I_n or just I, is the $n \times n$ diagonal matrix in which all the diagonal entries are equal to 1. $\quad I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
- Two matrices A and B are called equal if:
  - they have the same size, and
  - the same entries at the corresponding locations.

Example

$$\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \quad \text{but} \quad \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \neq \begin{bmatrix} -1 & 3 \\ 1 & 2 \end{bmatrix}$$

Also note that: $\begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 2 & 5 \end{bmatrix} \neq \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 5 \end{bmatrix}$

Matrix Addition

For any two $m \times n$ matrices A and B, the matrix $A + B$ is an $m \times n$ matrix whose (i, j)-entry is equal to the sum of (i, j)-entries of A and B.

$$A = [a_{ij}] \text{ and } B = [b_{ij}] \implies A + B = [a_{ij} + b_{ij}]$$

Example. If $A = \begin{bmatrix} 2 & -3 & 3 \\ 3 & -2 & -\frac{1}{3} \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & \frac{2}{3} \end{bmatrix}$ Then

$$A + B = \begin{bmatrix} 2 & -3 & 3 \\ 3 & -2 & -\frac{1}{3} \end{bmatrix} + \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & \frac{2}{3} \end{bmatrix} = \begin{bmatrix} 3 & -2 & 3 \\ 2 & -1 & \frac{1}{3} \end{bmatrix}$$

Example. Can we add $A = \begin{bmatrix} 2 & -3 & 3 \\ 3 & -2 & -\frac{1}{3} \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 0 & -1 \end{bmatrix}$?

No, because they are not of the same size.

Scalar multiples of a matrix

Just like vectors, we can multiply a matrix by a scalar :

If $A = [a_{ij}]$ and c is a scalar then $cA = [ca_{ij}]$.

Example

If $A = \begin{bmatrix} 2 & -3 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ then :

$(-3)A = \begin{bmatrix} -6 & 9 & -3 \\ 0 & -3 & 3 \end{bmatrix}$ and $0A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

What is $-A$? It is just $(-1)A = \begin{bmatrix} -2 & 3 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

One can also consider more complicated expressions :

Example

If $A = \begin{bmatrix} 2 & -3 & 3 \\ 3 & -2 & -\frac{1}{3} \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & \frac{2}{3} \end{bmatrix}$

Then:

$$\begin{aligned} 3A - 2B &= 3\begin{bmatrix} 2 & -3 & 3 \\ 3 & -2 & -\frac{1}{3} \end{bmatrix} - 2\begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & \frac{2}{3} \end{bmatrix} \\ &= \begin{bmatrix} 6 & -9 & 9 \\ 9 & -6 & -1 \end{bmatrix} - \begin{bmatrix} 2 & 2 & 0 \\ -2 & 2 & \frac{4}{3} \end{bmatrix} \\ &= \begin{bmatrix} 4 & -11 & 9 \\ 11 & -8 & -\frac{7}{3} \end{bmatrix} \end{aligned}$$

Basic properties of matrix addition/scalar multiplication

Suppose A, B, C are $m \times n$ matrices and r, s are scalars. Then :

- $A + B = B + A$
- $(A + B) + C = A + (B + C)$
- $A + 0_{m\times n} = 0_{m\times n} + A = A$ (Here $0_{m\times n}$ is the $m \times n$ zero matrix.)
- $r(A + B) = rA + rB$
- $(r + s)A = rA + sA$
- $r(sA) = (rs)A$

Product of a matrix and a vector

Given an $m \times n$ matrix A and an $n \times 1$ vector $\mathbf{v}$, the product $A\mathbf{v}$ was defined as follows :

$$A\mathbf{v} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

So, the result is an $m \times 1$ vector.

$$\text{product of} \quad m \times n \quad \text{and} \quad n \times 1 \quad \rightsquigarrow \quad m \times 1$$

Example.

$$\begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & -1 & -2 & 1 \\ 2 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -3 \\ -1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} + (-3) \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ 4 \\ -1 \end{bmatrix}$$

Matrix Multiplication

We want to define the product of two matrices.

- We have just defined the product of an $m \times n$ matrix and an $n \times 1$ matrix.

- If A is an $m \times n$ matrix and B is an $n \times p$ matrix, then the product AB is defined and is an $m \times p$ matrix.
- Write B as
$$B = [\, \mathbf{b}_1 \; \mathbf{b}_2 \; \cdots \; \mathbf{b}_p \,]$$
where $\mathbf{b}_1, ..., \mathbf{b}_p$ are $n \times 1$ vectors.
- We define
$$AB = [\, A\mathbf{b}_1 \;\; A\mathbf{b}_2 \;\; \cdots \;\; A\mathbf{b}_p \,]$$

$$m \times n \qquad n \times p \qquad \rightsquigarrow \qquad m \times p$$

Matrix Multiplication

$$B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_p] \quad \Rightarrow \quad AB = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_p]$$

Example

$$\begin{bmatrix} 1 & -3 & 4 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 5 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 0 \\ -3 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} 1 & -3 & 4 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 5 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} & \begin{bmatrix} 1 & -3 & 4 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 5 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ \frac{1}{2} \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} -13 & 1 \\ 1 & -\frac{3}{2} \\ 7 & -1 \\ 8 & -\frac{1}{2} \end{bmatrix}$$

$4 \times 3 \qquad 3 \times 2 \quad \rightsquigarrow \quad 4 \times 2$

Matrix Multiplication

$$B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_p] \quad \Rightarrow \quad AB = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_p]$$

Example

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & 5 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 1 & 5 \\ 3 & -1 & 2 \\ 2 & -\frac{1}{2} & 4 \end{bmatrix} = \left[\begin{bmatrix} 1 & -1 & 2 \\ 2 & 5 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ 3 \\ 2 \end{bmatrix} \ \begin{bmatrix} 1 & -1 & 2 \\ 2 & 5 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ -\frac{1}{2} \end{bmatrix} \ \begin{bmatrix} 1 & -1 & 2 \\ 2 & 5 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \\ 4 \end{bmatrix} \right]$$

$$= \begin{bmatrix} 1 \times \frac{1}{2} + (-1) \times 3 + 2 \times 2 & 1 \times 1 + (-1) \times (-1) + 2 \times (-\frac{1}{2}) & 1 \times 5 + (-1) \times 2 + 2 \times 4 \\ 2 \times \frac{1}{2} + 5 \times 3 + 0 \times 2 & 2 \times 1 + 5 \times (-1) + 0 \times (-\frac{1}{2}) & 2 \times 5 + 5 \times 2 + 0 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{2} & 1 & 11 \\ 16 & -3 & 20 \end{bmatrix}$$

Example

The product $\begin{bmatrix} 1 & -3 & 4 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 5 & -1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 0 \\ -1 & 0 & 1 \\ 1 & 2 & -1 \\ 0 & 1 & -1 \end{bmatrix}$ is not defined.

$$4 \times 3 \qquad 4 \times 3 \rightsquigarrow \text{☹}$$

• The number of columns of the first matrix is NOT equal to the number of rows of the second matrix.

Remember the Rule :

$$m \times n \quad n \times p \quad \rightsquigarrow \quad m \times p$$

Example.

- If A is 7×4 and B is 4×3 then AB is defined and is a 7×3 matrix.
- However, if A is 3×2 and B is 5×2 then AB is not defined!

Another method of calculation of AB

Product of $m \times n$ and $n \times p$

- The (i,j)-entry of AB is the product of the i-th row of A with the j-th column of B.

$$AB = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \vdots \\ \mathbf{r}_m \end{bmatrix} \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_p \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 \cdot \mathbf{c}_1 & \mathbf{r}_1 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_1 \cdot \mathbf{c}_p \\ \mathbf{r}_2 \cdot \mathbf{c}_1 & \mathbf{r}_2 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_2 \cdot \mathbf{c}_p \\ \vdots & \vdots & & \vdots \\ \mathbf{r}_m \cdot \mathbf{c}_1 & \mathbf{r}_m \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_m \cdot \mathbf{c}_p \end{bmatrix}$$

- In other words, if $A = [a_{ij}]$ is $m \times n$ and $B = [b_{ij}]$ is $n \times p$ then the (i,j)-entry of AB is

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

$$\begin{bmatrix} 2 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 2(0) + (-1)4 & 2(1) + (-1)(1) \\ (-2)(0) + 3(4) & (-2)(1) + 3(1) \end{bmatrix} = \begin{bmatrix} -4 & 1 \\ 12 & 1 \end{bmatrix}$$

Another method of calculation of AB

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- The (i,j)-entry of AB is the product of the i-th row of A with the j-th column of B.

$$AB = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \vdots \\ \mathbf{r}_m \end{bmatrix} \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_p \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 \cdot \mathbf{c}_1 & \mathbf{r}_1 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_1 \cdot \mathbf{c}_p \\ \mathbf{r}_2 \cdot \mathbf{c}_1 & \mathbf{r}_2 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_2 \cdot \mathbf{c}_p \\ \vdots & \vdots & & \vdots \\ \mathbf{r}_m \cdot \mathbf{c}_1 & \mathbf{r}_m \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_m \cdot \mathbf{c}_p \end{bmatrix}$$

- In other words, if $A = [a_{ij}]$ is $m \times n$ and $B = [b_{ij}]$ is $n \times p$ then the (i,j)-entry of AB is

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

$$\begin{bmatrix} 3 & -1 \\ 2 & 5 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 & 5 \\ -2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} (3)(0) + (-1)(-2) & (3)(-1) + (-1)(1) & (3)(5) + (-1)(1) \\ (2)(0) + (5)(-2) & (2)(-1) + (5)(1) & (2)(5) + (5)(1) \\ (-1)(0) + (0)(-2) & (-1)(-1) + (0)(1) & (-1)(5) + (0)(1) \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -4 & 14 \\ -10 & 3 & 15 \\ 0 & 1 & -5 \end{bmatrix}$$

Example

$$\begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \end{bmatrix} \begin{bmatrix} -2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} -2 & 3 & 4 \\ 4 & -6 & -8 \\ 0 & 0 & 0 \\ -6 & 9 & 12 \end{bmatrix}$$

However,

$$\begin{bmatrix} -2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \end{bmatrix} = ☹$$

Some Warnings

- Sometimes AB is defined, but BA is not.

  For example, if A is 3×2 and B is 2×4, then AB is defined (and it is 3×4) but BA is not defined.

$$\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}\begin{bmatrix} * & * & * & * \\ * & * & * & * \end{bmatrix} \text{ vs. } \begin{bmatrix} * & * & * & * \\ * & * & * & * \end{bmatrix}\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}$$

- Sometimes AB and BA are defined but have different sizes.

  For example, if A is 3×2 and B is 2×3 then AB is 3×3 whereas BA is 2×2, so $AB \neq BA$.

$$\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}\begin{bmatrix} * & * & * \\ * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \text{ whereas } \begin{bmatrix} * & * & * \\ * & * & * \end{bmatrix}\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix} = \begin{bmatrix} * & * \\ * & * \end{bmatrix}$$

Some Warnings (cont.)

- Even if both AB and BA are defined and have the same size, we might have $AB \neq BA$.

Example

If $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

then

$$AB = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$$

but

$$BA = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

so

$$AB \neq BA.$$

Summary of today

- If A is an $m \times n$ matrix and B is an $n \times p$ matrix, then the product AB can be defined and is an $m \times p$ matrix.

$$m \times n \qquad n \times p \quad \rightsquigarrow \quad m \times p$$

- The (i, j)-entry of AB is equal to the product of i-th row of A with the j-th column of B.

$$AB = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \vdots \\ \mathbf{r}_m \end{bmatrix} \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_p \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 \cdot \mathbf{c}_1 & \mathbf{r}_1 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_1 \cdot \mathbf{c}_p \\ \mathbf{r}_2 \cdot \mathbf{c}_1 & \mathbf{r}_2 \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_2 \cdot \mathbf{c}_p \\ \vdots & \vdots & & \vdots \\ \mathbf{r}_m \cdot \mathbf{c}_1 & \mathbf{r}_m \cdot \mathbf{c}_2 & \cdots & \mathbf{r}_m \cdot \mathbf{c}_p \end{bmatrix}$$

- Another way to compute AB is using the following relation:

$$B = [\mathbf{c}_1 \;\; \mathbf{c}_2 \;\; \cdots \;\; \mathbf{c}_p] \quad \Rightarrow \quad AB = [A\mathbf{c}_1 \;\; A\mathbf{c}_2 \;\; \cdots \;\; A\mathbf{c}_p]$$